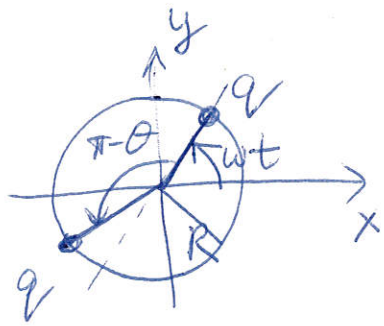
P1



$$\vec{d} = \sum_{j=1,2} q_j \vec{r}_j = qR\Big[\cos(\omega t)\hat{x} + \sin(\omega t)\hat{y} + \cos(\omega t + \pi - \theta)\hat{x} + \sin(\omega t + \pi - \theta)\hat{y}\Big]$$

(a)

$$\vec{d} = qR\Big\{\underbrace{\big[\cos(\omega t) - \cos(\omega t - \theta)\big]}_{\cos(\omega t)(1-\cos\theta) - \sin(\omega t)\sin\theta}\hat{x} + \underbrace{\big[\sin(\omega t) - \sin(\omega t - \theta)\big]}_{\sin(\omega t)(1-\cos\theta) + \cos(\omega t)\sin\theta}\hat{y}\Big\}$$

In order for electric dipole contribution to be as small as electric quadrupole, the angle $\theta \ll 1$
(for $\theta \sim 1$ dipole intensity is much stronger than quadrupole)
$\Rightarrow$ can approximate $\cos\theta \simeq 1$, $\sin\theta \simeq \theta$

$$\Rightarrow \vec{d} \cong qR\theta\Big\{-\sin(\omega t)\hat{x} + \cos(\omega t)\hat{y}\Big\}$$

(check: for $\theta = 0$ $\vec{d} = 0$)

To obtain $Q_{\alpha\beta}$ we can set $\theta = 0$ as $Q \neq 0$ when $\theta = 0$.
This calculation was done previously. The result

$$\hat{Q} = 2qR^2 \begin{pmatrix} 3\cos^2\omega t - 1 & 3\cos\omega t \sin\omega t & 0 \\ 3\cos\omega t \sin\omega t & 3\sin^2\omega t - 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\vec{Q}_\alpha = Q_{\alpha\beta} n_\beta \Rightarrow Q_x = 2qR^2\Big((3\cos^2\omega t - 1)n_x + 3\cos\omega t \sin\omega t\, n_y\Big)$$

$$Q_y = 2qR^2\Big((3\sin^2\omega t - 1)n_y + 3\cos\omega t \sin\omega t\, n_x\Big)$$

$$Q_z = -2qR^2 n_z$$

$$\dddot{\hat{Q}} = \frac{d^3}{dt^3}\left\{ 3qR^2 \begin{pmatrix} \cos(2\omega t) & \sin(2\omega t) & 0 \\ \sin(2\omega t) & -\cos(2\omega t) & 0 \\ 0 & 0 & 0 \end{pmatrix} + \text{const} \right\} =$$

$$= 3qR^2 (2\omega)^3 \begin{pmatrix} \sin(2\omega t) & -\cos(2\omega t) & 0 \\ -\cos(2\omega t) & -\sin(2\omega t) & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\dddot{Q}_x = (24 q R^2 \omega^3)\left(\sin(2\omega t)\, n_x - \cos(2\omega t)\, n_y\right)$$

$$\dddot{Q}_y = (24 q R^2 \omega^3)\left(-\cos(2\omega t)\, n_x - \sin(2\omega t)\, n_y\right)$$

From lectures (

$$p^{(EQ)} = \frac{\mu_0}{720\pi c^3} \sum_{\alpha\beta} \dddot{Q}_{\alpha\beta}^2 = \frac{\mu_0}{720\pi c^3}\, 2(24 q R^2 \omega^3)^2 = \frac{8\mu_0 q^2 R^4 \omega^6}{5\pi c^3}$$

$$\hookrightarrow \dddot{Q}_{xx}^2 + \dddot{Q}_{xy}^2 + \dddot{Q}_{yx}^2 + \dddot{Q}_{yy}^2 =$$

$$= (24 q R^2 \omega^3)^2 (2)\left(\sin^2(2\omega t) + \cos^2(2\omega t)\right)$$

Now electric dipole $\ddot{\vec{d}} = qR\theta\omega^2\left(\sin(\omega t)\hat{x} - \cos(\omega t)\hat{y}\right)$

$$p^{(ED)} = \frac{\mu_0}{6\pi c} |\ddot{\vec{d}}|^2 = \frac{\mu_0}{6\pi c}(qR\theta\omega^2)^2 = \frac{\mu_0 q^2 R^2 \omega^4 \theta^2}{6\pi c}$$

$$p^{(ED)} = p^{(EQ)} \Rightarrow \frac{\mu_0 q^2 R^2 \omega^4 \theta^2}{6\pi c} = \frac{8\mu_0 q^2 R^4 \omega^6}{5\pi c^3}$$

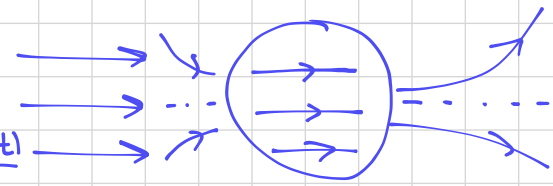
$$\theta^2 = \frac{48}{5}\frac{R^2\omega^2}{c^2}$$

$$\theta = 2\sqrt{\frac{12}{5}}\frac{R\omega}{c} \ll 1 \text{ for } R\omega \ll c.$$

Metal sphere in a homogeneous external <u>ac</u> field

$$\vec{E}_0(t) = \vec{E}_0 \cos\omega t = \frac{1}{2}\vec{E}_0 e^{i\omega t} + \frac{1}{2}\vec{E}_0 e^{-i\omega t}$$

(a) Total field outside:



$$\vec{E}_{out}(t) = \vec{E}_0(t) + A\underbrace{\frac{3\hat{r}(\hat{r}\cdot\vec{E}_0(t)) - \vec{E}_0(t)}{r^3}}_{\text{field of a dipole}}$$

Total field inside: $\vec{E}_{in}(t) = B\,\vec{E}_0(t)$ expect it to be less than $\vec{E}_0(t)$ due to polarization $\vec{P}(t)$.

Boundary conditions: 1) $\vec{D}_{in}\cdot\hat{r}\big|_{r=R} = \vec{D}_{out}\cdot\hat{r}\big|_{r=R}$

$$\Rightarrow \varepsilon B\,\vec{E}_0\cdot\hat{r} = \varepsilon_0\left(\vec{E}_0\cdot\hat{r} + A\,\frac{3\hat{r}\cdot\vec{E}_0 - \vec{E}_0\cdot\hat{r}}{R^3}\right) =$$

$$= \varepsilon_0\,\vec{E}_0\cdot\hat{r}\left(1 + \frac{2A}{R^3}\right).$$

$$\frac{\varepsilon}{\varepsilon_0}B = 1 + \frac{2A}{R^3}.$$

2) $\vec{E}_{in}(t)\times\hat{r}\big|_{r=R} = \vec{E}_{out}\times\hat{r}\big|_{r=R} \Rightarrow B\,\vec{E}_0\times\hat{r} = \vec{E}_0\times\hat{r} - \frac{A}{R^3}\vec{E}_0\times\hat{r}$

$$B = 1 - \frac{A}{R^3}.$$

Thus we find $\left(\frac{\varepsilon}{\varepsilon_0}+2\right)B = 3$, $B = \dfrac{3}{2+\frac{\varepsilon}{\varepsilon_0}} \Rightarrow \dfrac{A}{R^3} = 1 - B = \dfrac{\frac{\varepsilon}{\varepsilon_0}-1}{\frac{\varepsilon}{\varepsilon_0}+2}$.

$$\vec{E}_{in} = \frac{3}{2+\varepsilon/\varepsilon_0}\vec{E}_0.$$

For time-dependent field: $\vec{E}_{in}(t) = \dfrac{3}{2+\frac{\varepsilon(\omega)}{\varepsilon_0}}\frac{1}{2}\vec{E}_0 e^{-i\omega t} + \dfrac{3}{2+\frac{\varepsilon(-\omega)}{\varepsilon_0}}\frac{1}{2}\vec{E}_0 e^{i\omega t}$

$$= \frac{3/2}{2+\frac{\varepsilon(\omega)}{\varepsilon_0}}\vec{E}_0 e^{-i\omega t} + \underbrace{c.c.}_{\text{complex conjugate}}$$

Analytic property of $\varepsilon(\omega)$: $\varepsilon(-\omega) = \varepsilon^*(\omega)$.

Polarization $\vec{P} = \vec{D}_{in} - \varepsilon_0\vec{E}_{in} = (\varepsilon-\varepsilon_0)\vec{E}_{in} = (\varepsilon(\omega)-\varepsilon_0)\dfrac{3/2}{2+\frac{\varepsilon(\omega)}{\varepsilon_0}}\vec{E}_0 e^{-i\omega t} +$

$$+ (\varepsilon^*(\omega)-\varepsilon_0)\frac{3/2}{2+\frac{\varepsilon^*(\omega)}{\varepsilon_0}}\vec{E}_0 e^{i\omega t}$$

$$\vec{P}(t) = \frac{3\varepsilon_0(\varepsilon(\omega)-\varepsilon_0)}{2\varepsilon_0+\varepsilon(\omega)}\cdot\frac{1}{2}\vec{E}_0 e^{-i\omega t} + \frac{3\varepsilon_0(\varepsilon^*(\omega)-\varepsilon_0)}{2\varepsilon_0+\varepsilon^*(\omega)}\cdot\frac{1}{2}\vec{E}_0 e^{+i\omega t}$$

The current density $\vec{j}(t) = \dfrac{\partial\vec{P}}{\partial t}$

$$\vec{j}(t) = \frac{-3i\omega\varepsilon_0[\varepsilon(\omega)-\varepsilon_0]}{2\varepsilon_0+\varepsilon(\omega)}\,\frac{1}{2}\vec{E}_0 e^{-i\omega t} + \frac{3i\omega\varepsilon_0[\varepsilon^*(\omega)-\varepsilon_0]}{2\varepsilon_0+\varepsilon^*(\omega)}\cdot\frac{1}{2}\vec{E}_0 e^{i\omega t}$$

Can write the same in terms of electric susceptibility $\chi(\omega)$

$$\chi(\omega) = \frac{\varepsilon(\omega)-\varepsilon_0}{\varepsilon_0}, \quad \varepsilon(\omega) = \varepsilon_0 + \varepsilon_0\chi(\omega)$$

$$\vec{p}(t) = \frac{3\varepsilon_0\chi(\omega)}{3+\chi(\omega)}\cdot\frac{1}{2}\vec{E}_0 e^{-i\omega t} + \frac{3\varepsilon_0\chi^*(\omega)}{3+\chi^*(\omega)}\cdot\frac{1}{2}\vec{E}_0 e^{i\omega t}$$

$$\vec{j}(t) = \frac{-i\omega\varepsilon_0\chi(\omega)}{1+\frac{\chi(\omega)}{3}}\cdot\frac{1}{2}\vec{E}_0 e^{-i\omega t} + \frac{i\omega\varepsilon_0\chi^*(\omega)}{1+\frac{\chi^*(\omega)}{3}}\cdot\frac{1}{2}\vec{E}_0 e^{i\omega t}$$

Drude's model: $\varepsilon(\omega) = \varepsilon_0\left(1-\frac{\omega_p^2}{\omega(\omega+i\gamma)}\right) \Rightarrow \chi(\omega) = -\frac{\omega_p^2}{\omega(\omega+i\gamma)}$.

(b) Here we set $\gamma=0 \Rightarrow \chi(\omega) = -\frac{\omega_p^2}{\omega^2}$.

$\vec{p}$ and $\vec{j}$ have common denominator $1+\frac{1}{3}\chi(\omega) = 1-\frac{\omega_p^2}{3\omega^2}$.

Therefore they reach maximum values at <u>the resonance</u>

when $\frac{\omega_p^2}{3\Omega^2} = 1 \Rightarrow$ resonance frequency $\Omega = \frac{\omega_p}{\sqrt{3}}$.

The resonance is between external driving field $\vec{E}(t)$ and plasma oscillations within the spherical conductor.

(e) Joule's heating, per unit volume of the sphere,

$$P = \overline{\vec{j}(t)\cdot\vec{E}_1(t)} = \int_0^{2\pi/\omega} dt\left(\frac{\omega}{2\pi}\right)\vec{j}(t)\cdot\vec{E}(t).$$

$$P = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} dt\left(\vec{j}_\omega e^{-i\omega t} + \vec{j}_\omega^* e^{i\omega t}\right)\cdot\left(\vec{E}_{in}(\omega)e^{-i\omega t} + \vec{E}_{in}^*(\omega)e^{i\omega t}\right) =$$

$$= \vec{j}_\omega\cdot\vec{E}_{in}^*(\omega) + \vec{j}_\omega^*\cdot\vec{E}_{in}(\omega).$$

$$P = \frac{i\omega\varepsilon_0\chi^*(\omega)}{1+\frac{\chi^*(\omega)}{3}}\,\frac{1}{2}\vec{E}_0\cdot\frac{3}{3+\chi(\omega)}\,\frac{1}{2}\vec{E}_0 + c.c. =$$

$$= \frac{\varepsilon_0}{4}\vec{E}_0^2\left(\frac{i\omega\chi^*(\omega)}{1+\frac{\chi^*(\omega)}{3}}\,\frac{1}{1+\frac{\chi(\omega)}{3}} + \frac{-i\omega\chi(\omega)}{1+\frac{\chi(\omega)}{3}}\,\frac{1}{1+\frac{\chi^*(\omega)}{3}}\right)$$

where now $\chi(\omega) = -\frac{\omega_p^2}{\omega(\omega+i\gamma)} = \underbrace{\chi'(\omega)}_{\text{real}} + i\,\underbrace{\chi''(\omega)}_{\text{imaginary parts}}$.

Thus $P = \frac{\varepsilon_0}{4} \vec{E}_0^2 \frac{i\omega(\chi^* - \chi)}{\left(1+\frac{\chi}{3}\right)\left(1+\frac{\chi^*}{3}\right)} =$ Lorentzian shape!

$$= \frac{\varepsilon_0}{4} \vec{E}_0^2 \frac{i\omega\left(-2i\,\chi''(\omega)\right)}{\left(1+\frac{\chi'(\omega)}{3}\right)^2 + \left(\frac{\chi''(\omega)}{3}\right)^2} = \frac{\varepsilon_0}{2} |\vec{E}_0|^2 \frac{\omega\chi''(\omega)}{\left(1+\frac{\chi'(\omega)}{3}\right)^2 + \left(\frac{\chi''(\omega)}{3}\right)^2}$$

$$\chi(\omega) = -\frac{\omega_p^2(\omega - i\gamma)}{\omega(\omega^2+\gamma^2)} \Rightarrow \begin{cases} \chi'(\omega) = -\frac{\omega_p^2}{\omega^2+\gamma^2} \approx -\frac{\omega_p^2}{\omega^2} \\ \chi''(\omega) = \frac{\omega_p^2\gamma}{\omega(\omega^2+\gamma^2)} \approx \frac{\omega_p^2\gamma}{\omega^3} \end{cases}$$

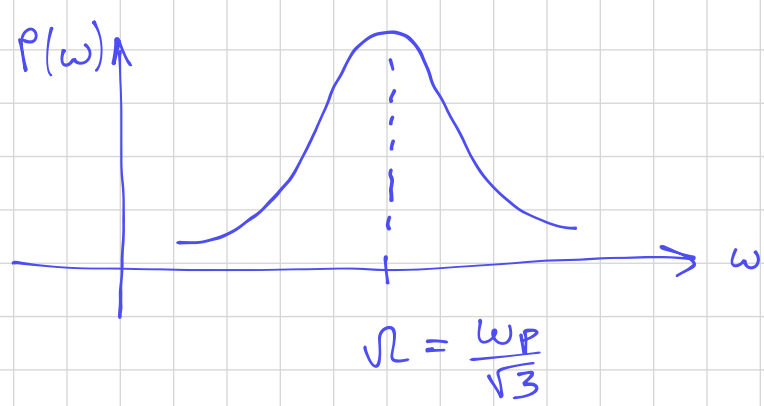
assuming $\gamma \ll \omega$ as problem tells us.

$$P = \frac{\frac{1}{2}\varepsilon_0 |\vec{E}_0|^2 \, \omega \frac{\omega_p^2\gamma}{\omega^3}}{\left(1-\frac{\omega_p^2}{3\omega^2}\right)^2 + \left(\frac{\omega_p^2\gamma}{3\omega^3}\right)^2}.$$

Near the resonance $\omega \approx \Omega = \frac{\omega_p}{\sqrt{3}} \Rightarrow$ can replace ω by Ω in all terms except $\left(1-\frac{\omega_p^2}{3\omega^2}\right)$ (as this would just nullify it).

$$P = \frac{1}{2}\varepsilon_0 |\vec{E}_0|^2 \frac{\gamma \frac{\omega_p^2}{\Omega^2}}{\left(1-\frac{\omega_p^2}{3\omega^2}\right)^2 + \left(\frac{\gamma\omega_p^2}{3\Omega^3}\right)^2} = \frac{1}{2}\varepsilon_0 |\vec{E}_0|^2 \frac{3\gamma}{\left(1-\frac{\omega_p^2}{3\omega^2}\right)^2 + \frac{3\gamma^2}{\omega_p^2}}.$$

Use $\frac{\omega_p^2}{\Omega^2} = 3$



Energy absorption (heating) reaches maximum at the resonance when $\omega = \Omega = \omega_p/\sqrt{3}$. The rate of absorption is controlled by $\gamma = \frac{1}{\tau}$.